



SkiLLMo: Normalized ESCO Skill Extraction through Transformer Models

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Abstract

In recent years, natural language processing (NLP) technologies have made a significant contribution in addressing a number of labour market tasks. One of the most interesting challenges is the automatic extraction of competences from unstructured texts.

This paper presents a pipeline for efficiently extracting and standardizing skills from job advertisements using NLP techniques. The proposed methodology leverages open-source Transformer and Large Language Models to extract skills and map them to the European labour market taxonomy, ESCO.

To address the computational challenges of processing lengthy job advertisements, a BERT model was fine-tuned to identify text segments likely containing skills. This filtering step reduces noise and ensures that only relevant content is processed further. The filtered text is then passed to an LLM, which extracts implicit and explicit hard and soft skills through prompt engineering. The extracted skills are subsequently matched with entries in a vector store containing the ESCO taxonomy to achieve standardization.

Evaluation by domain experts shows that the pipeline achieves a precision of 91% for skill extraction, 80% for skill standardization and a combined overall precision of 79%. These results demonstrate the effectiveness of the proposed approach in facilitating structured and standardized skill extraction from job postings.

CCS Concepts

• Computing methodologies → Information extraction; Neural networks; • Applied computing → Business intelligence.

Keywords

Skill Extraction, Large Language Models, Transformer Models, Information Extraction, Labor Market

ACM Reference Format:

Lorenzo Malandri, Fabio Mercorio, and Antonio Serino. 2025. SkiLLMo: Normalized ESCO Skill Extraction through Transformer Models. In *The 40th ACM/SIGAPP Symposium on Applied Computing (SAC '25)*, March 31–April 4, 2025, Catania, Italy. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3672608.3707960>

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ACM ISBN 979-8-4007-0629-5/25/03
<https://doi.org/10.1145/3672608.3707960>

1 Introduction

In the global economy, the labor market serves as the domain where the interplay between employer demand and applicant supply is characterized, forming a dense and intense system for the exchange of human resources. Over time, a series of social dynamics has caused this area to evolve more and more into the infosphere: the advent of job search platforms, such as LinkedIn or Xing, has brought a significant portion of the labour market online, enabling the generation and analysis of Labour Market Information. Digital processing of data, including job adverts and CVs, enables the opportunity to collect, analyse and interpret a wide range of labour market data [14]. The analysis of large amounts of textual data allows the extraction of structured information from unstructured data. Skill extraction is still an open challenge that plays a key role in the labour market [11, 27]. This task carries substantial importance as its successful completion would facilitate valuable analyses for various stakeholders in the sector. For instance, the effective extraction of skills from CVs could enhance the efficiency of selection processes. Furthermore, analyzing the trends of the skills most in demand in job advertisements could provide valuable insights for training institutions to better tailor their curricula.

Natural Language Processing (NLP) plays a key role in this landscape. One of the main tasks in this domain is extracting structured information from unstructured text. However, supervised methods for extracting skills from unstructured text may face limitations due to the continuous need of updated skill labels. In the absence of representativeness in the training datasets, these new skills become impossible to identify.

These tasks rely heavily on the availability of OJA datasets that are properly labeled for training machine learning models and assessing their performance. While some datasets have been introduced in previous research, very few are publicly accessible, and those that are often suffer from limited scope in terms of skill and occupation coverage. Additionally, none of these datasets include clear annotation guidelines, which makes it difficult to evaluate the consistency of the data [32]. The absence of a comprehensive benchmark introduces two key limitations: it hampers the ability to compare results across studies and poses challenges in developing robust models due to the lack of consistent training data.

Large Language Models (LLMs) have revolutionised NLP outperforming State Of The Art (SOTA) technologies in multiple decision making tasks [7, 9]. Thanks to their zero-shot and few-shots capabilities those models are able to perform tasks for which they were not trained. Combining Encoder-Only and Decoder-Only Transformer models we develop a methodology that allow to extract Skill

from Online Job Advertisements (OJA) and standardize them on the ESCO taxonomy.

Moreover, it is worth to notice that in such activities, usually referring to the tasks of Named Entity Recognition (NER) or Information Extraction (IE), structured information like taxonomies plays a crucial role in accurately identifying and categorizing relevant elements within unstructured text [21]. In order to tackle all the tasks described, we need a tool that represents and organizes the skills and competences in an organized manner, providing a *lingua franca* for the labour market. ESCO is the multilingual European classification of Skills, Competences and Occupations, which aims to create a more knowledgeable and aligned European labour market by improving the match between labour demand and supply on the European labour market, facilitating labour mobility and human resource management on a transnational level, providing useful information to training providers, and keeping up-to-date through Data Driven approaches.

1.1 Contributions

The contributions of this work are three-fold:

- (1) We fine-tune a BERT model to filter OJA, keeping the text windows with a higher probability of containing skills, in order to reduce noise and computational burden;
- (2) We develop a Skill Extraction and ESCO Standardization tools using an open source LLM and a Vector Database;
- (3) We conduct a comprehensive evaluation involving domain experts.

The remainder of this paper is organised as follows: chapter 2 introduces the technologies used in this work and illustrates the state of the art with regard to skill extraction methodologies, chapter 3 defines all the components that make up SkiLLMo, chapter 4 describes the characteristics of the implemented experiment, chapter 5 sets out and discusses the results, and finally, chapter 6 offers future directions with regard to our work.

2 Background and Related Works

2.1 Transformer Models

Transformer models [28] are a type of deep learning architecture that has made significant advancements in various other fields of machine learning, especially in NLP. The key innovation of Transformer models is the use of a self-attention mechanism, which allows them to capture dependencies between different words or elements in a sequence, regardless of their distance from each other. This self-attention mechanism allows Transformers to excel at handling long-range dependencies and has largely replaced recurrent neural networks (RNNs) and convolutional neural networks (CNNs) in many NLP tasks. The self-attention mechanism allows the model to weigh the importance of different words or elements in a sequence when processing each word, capturing contextual information effectively. This, among others, allows to generate vector representations of words contextualized in the sentences they belong to. In addition to the advantage of being able to realise multiple tasks through deep textual understanding, Transformer architectures can parallelize the computation of self-attention due to the

inherent structure of the architecture that does not require sequential dependencies between tokens. This has enabled training on huge amounts of data using devices with limited resources.

2.2 Large Language Models

LLMs are powerful computational models designed to understand and generate human-like text by harnessing vast amounts of textual data, and have taken Natural Language Processing (NLP) by storm, achieving state-of-the-art performance in many tasks [22]. Typically these models are based on Transformer architecture [28], powered by Attention mechanism [1, 20] and are composed by decoder-only stack. These models are initially trained on Autoregressive task [24], where given a sequence of words $S = (w_1, w_2, \dots, w_{n-1})$ the training objective is to maximize the log-likelihood $\sum_i \log P(w_i | w_1, w_2, \dots, w_{i-1}; \theta^T)$ where θ^T are the model parameters, in order to predict the next word in the sequence $\prod_{i=1}^n P(w_i | w_1, \dots, w_{i-1})$. After being pre-trained, these models are fine-tuned for several tasks, providing examples of Natural Language Inference [24]. Thanks to their capability to learn from context, known as in-context learning [25], LLMs can accomplish specific tasks with high accuracy [33], exploiting prompt engineering methodologies such as zero-shot [29] and few-shot learning [5].

2.3 Related Works on Skill Extraction

The literature on Skill Extraction encompasses various methodologies aimed at capturing relevant competencies from job-related texts. This section provides an overview of the major approaches: exact matching, fuzzy matching, topic modeling, deep neural network (DNN) approaches, transformer-based approaches, and mixed methods.

Exact Matching. Early approaches relied on exact matching techniques. Calanca et al. [6] introduced a semi-automatic method to detect soft skills in job ads using manual annotation, clustering with Word2Vec, and cosine similarity-based grouping. Gugnani et al. [13] developed an ontology-based method to classify skills in CVs, achieving 80% precision and 88% recall. Javed et al. [17] combined exact matching with MCMC disambiguation, constructing a skill taxonomy and achieving 90% precision and 73% recall.

Fuzzy Matching. Fuzzy matching was introduced to handle variability in skill terminologies. Boselli et al. [4] used fuzzy matching to compare skill expressions with ESCO taxonomy labels, using metrics like Levenshtein Distance and Jaccard Similarity. Skills were validated if the similarity exceeded 70%, providing flexibility in capturing skill variations.

Topic Modelling. Latent Dirichlet Allocation (LDA) has been applied to extract skills from job descriptions. Gurcan and Cagiltay [15] identified 48 skill clusters using LDA, while De Mauro et al. [8] defined a domain-specific vocabulary before applying LDA, resulting in nine distinct skill sets. Both approaches provided structured skill representations without standardization.

Deep Neural Network Approaches. DNNs enabled more nuanced skill extraction. Li et al. [19] used entity tagging and market-aware ranking to extract skills from job ads. Jiechieu and Tsope [18] framed skill extraction as a multi-label classification problem using

CNNs, achieving 91% precision and 99% recall. Goyal et al. [12] proposed JobXMLC, using a Graph Isomorphism Network to capture multi-level skill representations.

Transformer-Based Approaches. Transformer models marked a significant shift towards contextual understanding. Fang et al. [10] introduced RecruitPro, a domain-specific language model trained for recruitment tasks, achieving competitive recall and F1 scores. Zhang et al. [32] used BERT and SpanBERT with a Conditional Random Field layer to improve precision and recall. Bhola et al. [3] proposed BERT-XMLC for extreme multi-label classification, achieving a Mean Reciprocal Rank of 90%. Starting from resumes, Barducci et al. [2] fine-tuned a BERT model to perform NER achieving 62% and a recall of 60% in extracting hard skills and a precision of 21% and a recall of 28% in extracting soft skills.

Mixed Approaches. Mixed methodologies combined strengths from various techniques. Gugnani and Misra [14] integrated NER, part-of-speech tagging, and dictionary-based approaches, achieving 80% precision and 93% recall, demonstrating the benefits of hybrid strategies.

3 Defining SkiLLMo

In the field of skills recognition, current methodologies exhibit significant limitations. On top of our knowledge, we propose a novel approach to address the extraction of both explicit and implicit skills from job postings, leveraging the capabilities of LLMs in the absence of an annotated "Gold Standard" dataset. Studies [31] have shown that LLMs can effectively perform domain-specific information extraction.

The task is formulated as an Information extraction problem, aiming to identify both "soft" and "hard" skills. Figure 1 shows the proposed pipeline, that comprises three main steps: information filtering, skills extraction, and skills standardization.

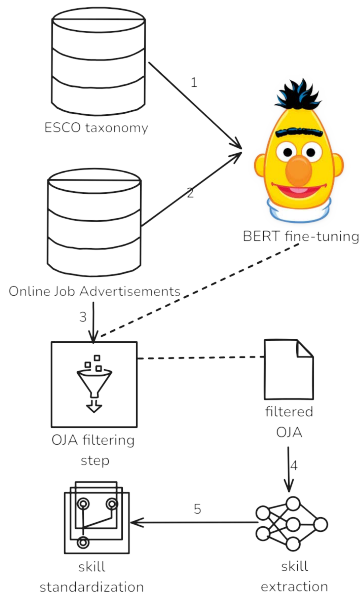


Figure 1: Proposed pipeline

3.1 Problem formulation

Given an OJA, the text undergoes cleaning and is input into a BERT-based binary classifier to estimate the probability of skills presence for each sentence, filtering relevant information.

The filtered OJA is then processed by an LLM, generating a list of skills, $Skill = (s_1, \dots, s_n)$, including both explicit and implicit skills. To obtain ESCO-standardized skills, each skill s_i is queried against a vector store containing ESCO skills, resulting in the most similar ESCO skill. Thus, the process yields a standardized list, $Skill_{ESCO} = (s_{1_{ESCO}}, \dots, s_{n_{ESCO}})$, for each job posting.

Algorithm 1: Entire process

Data: $OJA = (f_1, \dots, f_n)$
Result: $Skill_{ESCO} = (s_{1_{ESCO}}, \dots, s_{n_{ESCO}})$

```

FilteredOJA ← [];
for f in OJA do
    P(skill | f) ← BERT(f);
    Threshold ← 0.4;
    if P(skill | f) > Threshold then
        label ← 1;
        FilteredOJA.append(f);
    else
        label ← 0;
Skill ← LLM(prompt(FilteredOJA));
SkillESCO ← [];
for s in Skill do
    sESCO ← Argmax(query(s, DB));
    SkillESCO.append(sESCO);
return SkillESCO;
  
```

The methodologies for each step in the pipeline are detailed below.

3.2 OJA Filtering through BERT

Information filtering systems [16] automate the extraction of relevant information to handle large information flows and reduce overload. OJAs often contain extraneous elements from scraping processes, such as web page descriptions, layout, and menu strings, necessitating a cleaning step to reduce processing complexity and noise. We treat the OJA cleaning step as an information filtering task, considering an OJA as a set of sentences to be classify as relevance respect skills extraction task or not. Consider an OJA as a set of n sentences f_i .

$$OJA = (f_1, f_2, \dots, f_n)$$

the result of this step will be a set $FilteredOJA$ representing the set of m filtered sentences

$$FilteredOJA = \{c_1, c_2, \dots, c_m\} \text{ where } m \leq n$$

To perform this filtering step we fine-tune a *BERT base model* using the ESCO taxonomy, to identify skill-related sentences. The training dataset was constructed using descriptions of occupations and skills provided by ESCO. Specifically, the dataset included around 25,000

rows, with a balanced distribution between skill and occupation elements. Each row was labeled as either 0 (occupation) or 1 (skill). The rationale for using ESCO data is that the context in which a skill appears in job advertisements is often similar to its description in ESCO. By training BERT with both skill and occupation descriptions, the model can effectively learn to distinguish between these categories based on contextual clues. The model was trained on 80% of the dataset and validated on 20%. The model was then evaluated on a representative sample of 400 OJAs splitted in sentences and manually labelled by domain experts, achieving an accuracy of 40% and a recall of 81%. In order to improve those results we decide to fine-tune for a second time the *BERT model* using 300 hand labelled OJAs. The 75% of the dataset and validated on the remaining 25%, using a batch size of 16 for 4 epochs. Additionally, a cost matrix was introduced to give more relevance to the sentences containing skills, giving double weight to class 1. After the second fine-tuning the model achieve a precision of 71% and a recall of 93% evaluated on the remaining 100 hand labelled OJAs.

the fine-tuned model is available in the following HuggingFace repository: [serino28/BERT-OJA-SkillLess](https://huggingface.co/serino28/BERT-OJA-SkillLess)

3.3 Skill Extraction via LLM

Information extraction [23] is an NLP technique for identifying and extracting structured information from unstructured text, converting it into a form usable for analysis and research. Our approach treats skills extraction as an information extraction task, aiming to identify not only explicitly mentioned skills but also those implied through contextual information. As already specified we choose to use an LLM for information extraction capabilities in a specific domain, thanks to in-context learning.

Source Code 1: Prompt provided to the LLM

```

1  """This is a job advertisement: ``{filteredOJA}``
2  You have to analyse it and extract the labels of the
3  competences and of the skills in the advertisement.
4  You must extract complete and disambiguated recognised
5  skill-specific labels based on the context described in
6  the job advertisement.
7  You cannot extract half-assed skills, or unclear skills.
8  You must only return a list formed by the skills labels
9  recognised in the advertisement.
10 The list cannot present duplicates.
11 Here is an example of what the output should look like:
12 ["linear algebra", "fundamentals of physics",
13  "working in a team", "speak English"].
14 You cannot return anything other than the list.
15 If it is not a job advertisement, you must return an empty list."""

```

According to the *Open LLM Leaderboard*¹ we use the *solar-pro-Preview-instruct*, an instruction-tuned 22B parameters model that represents an optimal trade-off between size and performance.

In this step the model is therefore asked, via a prompt, to return the skills present within the filtered OJA, in the form of a list. Given a filtered OJA, consisting of a set of relevant sentences $FilteredOJA = \{c_1, \dots, c_m\}$,

$$LLM(FilteredOJA) \rightarrow Skill = \{s_1, \dots, s_k\}$$

The snippet 1 shows the prompt provided to the chosen model to perform skill extraction task. The model is asked to analyze the input filtered OJA and extract clear and complete skills based on the context of the input filtered OJA. Additionally, to automate the next step the model is asked to return skill label into square brackets, providing an example of the output shape. We also tried to input elements other than OJA and actually verified that the model returned an empty list.

3.4 ESCO Skill Standardization

The ESCO taxonomy aims to foster a more informed and harmonized European labor market by providing a reference framework for accurately defining the skills, abilities, and competencies required across different sectors. Despite the effectiveness of the prompt engineering approach employed earlier in achieving semantically detailed extractions, identifying exact terminology from the ESCO taxonomy remains a challenge. To address this, a Skill Standardization phase was implemented, enabling the mapping of model-extracted skills to their corresponding ESCO skills for each given Online Job Advertisement (OJA).

The development of the Skill Standardization step leveraged a Vector Database [26]. Such databases are specialized in storing, searching, and retrieving data in the form of multidimensional vectors. The primary purpose of a vector database is to enable efficient similarity-based search and retrieval, allowing the identification of elements closest to a query vector in a given dataset.

The Skill Standardization task was approached as a semantic retrieval problem: utilizing embedding vector representations to retrieve elements closest in the vector space, thereby retrieving skills semantically similar to a given query. In order to achieve this, a vector representation for both extracted skills and ESCO skills was required. This process necessitated choosing an appropriate embedding model and a Vector Store for storing vector representations of all skills in ESCO.

The embedding model used was *bge-base-en-v1.5* [30], a Transformer-based encoder-only known for strong performance in Natural Language Processing benchmarks according to the Massive Text Embedding Benchmark (MTEB) leaderboard², offering competitive performance with reduced computational requirements.

We use Chroma Vector Database³, which provides efficient tools for creating, storing, and searching embedding vectors. Chroma allows the creation of persistent data collections, reusable by specifying the embedding function and similarity function for semantic searches. Each element is indexed with a unique identifier, and metadata can be used to provide additional information.

A Vector Store was thus created, containing all embeddings for the ESCO skills. Since ESCO provides for each skill a preferred and alternative labels, within the vector database, for each ESCO skill, its own preferred label and all its alternative labels were represented, associating each of these with the preferred label as metadata.

¹https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard

²<https://huggingface.co/spaces/mteb/leaderboard>

³<https://docs.trychroma.com/>

Python (computer programming) = {
 Python (computer programming)
 Py3K
 Python
 Python2
 Python3
 Python prog

Source Code 2: ESCO skill storing example

```

1 collection.add(
2     documents=[
3         "Python (computer programming)",
4         "Py3K",
5         "Python",
6         "Python2",
7         "Python3",
8         "Python prog"
9     ],
10    metadatas=[
11        "Python (computer programming)",
12        "Python (computer programming)",
13        "Python (computer programming)",
14        "Python (computer programming)",
15        "Python (computer programming)"
16    ]
17 )

```

Code 2 shows a label representation example for the ESCO skill "Python (computer programming)".

Considering an LLM as non-deterministic technology which can extract the same skill, from different OJAs, as many different wordings, the representation of the same skill in many different syntactic captions allows for optimization of the skill standardization step.

Considering the list of skills $Skill = \{s_1, \dots, s_k\}$ extracted by the LLM, the skill standardization step involves, for each skill extracted from the model, making a query within the vector database representing ESCO skills. The query will return the preferred label (represented as metadata) of the ESCO element which will maximise the cosine similarity to the skill expressed as a query.

$$s_{ESCO_i} = \operatorname{argmax}_{DB_{skill}} (\cos(s_i, ESCO_{skill}))$$

$$Skill = (s_1, s_2, \dots, s_k)$$

⇓

$$Skill_{ESCO} = (s_{ESCO_1}, s_{ESCO_2}, \dots, s_{ESCO_k})$$

Iterating this process for each skill extracted by the LLM we obtain the list of ESCO skills present in the input filtered OJA.

4 Experimental Settings

In this section, we describe the experiment carried out using SkiLLMo, starting with a description of the data used, describing the characteristics of the experiment and describing the evaluation carried out.

4.1 Dataset Description

In 2019, Cedefop and Eurostat collaborated to establish the European Web Intelligence Hub ⁴ (WIH), a large-scale project that collects OJAs from over 700 sources across 27+1 EU countries. To date, the WIH has processed over 450 million unique job advertisements in 28 different languages, with the goal of incorporating OJAs into official labor statistics. The WIH applies ensemble learning methods to categorize job ads according to standardized taxonomies. The WIH NLP dataset is a representative sample of the broader European Web Intelligence Hub (WIH) dataset, stratified by country and occupation in accordance with the ESCO classification system. This dataset is available for research purposes through an agreement with Eurostat, and access is granted upon signing a non-disclosure agreement (NDA). Given its stratification by country and occupation according to the ESCO classification, along with the fact that it is specifically curated for research activities, the WIH NLP dataset is particularly well-suited for our purposes, offering a rich and structured source of labor market data for robust analysis.

4.2 Experiment Description

Since our experiment does not aim to evaluate the filtering created by BERT but rather the process of extracting and standardising the skills, we decide to use the 400 OJAs used previously for training and evaluating the OJA filtering step.

It is important to specify that the selected sample maintains the distribution of Job Occupations to the second digit represented in the original dataset. This therefore guarantees a true representation in terms of the distribution of job advertisements and does not allow for the extraction of only 'sectoral' skills, which could easily be the reason for strongly negative or positive performance.

In order to ensure determinism and reproducibility in the extraction phase, we set a temperature of 0 during experimentation.

The entire experiment was performed on a NVIDIA A100 80 GB, using vLLM⁵ for the inference.

4.3 Domain Experts Evaluation

Given the lack of a gold standard representing OJA and the respective ESCO skills within it, we decided to include 3 domain experts to conduct a human-based evaluation.

We divided the results to a skill level, creating a dataset representing an extracted and standardised skill for each row. We asked the domain experts to make a judgement of relevance with respect to the skill extracted by the LLM, evaluating the skill extraction phase and to express a degree of agreement between the extracted skill and the standardised ESCO skill.

Table 1 shows an example of the table structure provided to the domain experts.

The experts were asked to express a binary value between 0 and 1 for the relevance of the skill to the input-filtered OJA (0 if not relevant and 1 if relevant) and a categorical value from 1 to 5 (where 1 stands for strongly disagree and 5 stands for strongly agree) for the degree of agreement regarding the association between the skill extracted by the LLM and the ESCO skill.

⁴<https://www.cedefop.europa.eu/en/tools/skills-online-vacancies>

⁵<https://docs.vllm.ai/en/latest/>

Table 1: Dataset Example provided to domain experts

OJAid	LLMskill	ESCOskill	Job Ad	Relevance	Agreement
26459	Inventory management	Manage inventory	"We are looking for..."		
43891	Teamwork	Work in teams	"It also..."		
89568	SQL	SQL	"The Role..."		

5 Experimental Results

Skill Extraction results. Having three binary rates of relevance for each skill with respect to the input filtered OJA, we decided to take the mode between the three elements as the unique relevance indicator for each skill.

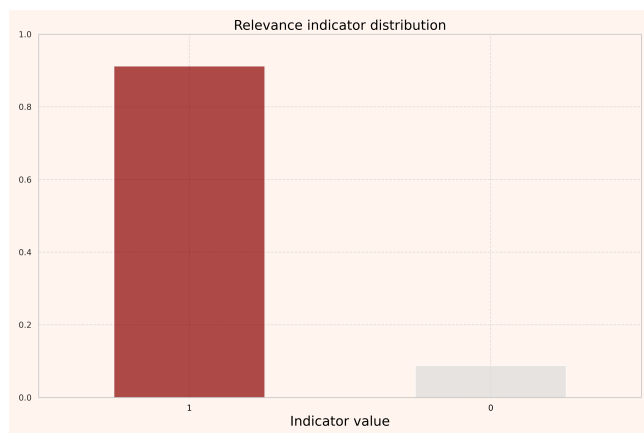


Figure 2: This figure shows the distribution of the unique relevance indicator values. Elements with a relevance value of 1 represent true positives, as they are extracted and present in the filtered OJA, while elements with a relevance value of 0 represent false positives, as they are extracted but not present.

Figure 2 shows the distribution of the unique relevance indicator. Elements with a relevance of 1 can be understood as true positives, as they are extracted and present in the filtered OJA, while elements with a relevance of 0 can be understood as false positives, as they are extracted but not present. This allows us to interpret this result as an accuracy of the skill extraction process of 91%.

Skill Standardization results. We decided to take the median of the three values expressed by the three domain experts as the single indicator for the agreement rate with respect to the association between the extracted skill and the standardised ESCO skill.

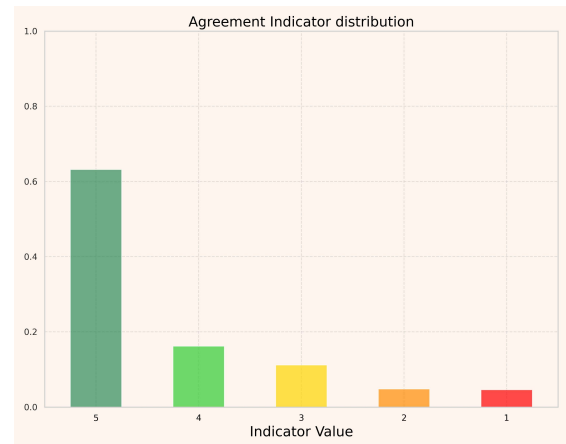


Figure 3: This figure displays the distribution of the unique agreement indicator values, ranging from 1 to 5. Values equal to 4 or 5 are considered correctly standardized, while values 3, 2, and 1 are classified as incorrectly standardized.

Figure 3 shows the distribution of the unique Agreement indicator. We chose to dichotomize this indicator by considering as correctly standardized those elements for which the unique indicator is equal to 4 or 5 and as incorrectly standardized those elements for which the unique indicator is equal to 3, 2, or 1. As with the relevance indicator, this dichotomization allows us to calculate a precision for the skill standardization step, which will be equal to the number of items with agreement indicator equal to 4 or 5. The standardization step totals an accuracy of 80%.

Overall results. Finally, we considered total precision by going to consider only those skills with a relevance indicator of 1 whose standardization has a degree of agreement of 4 or 5. This gave us insight into SkiLLMo performance in terms of the number of correct items extracted and standardized.

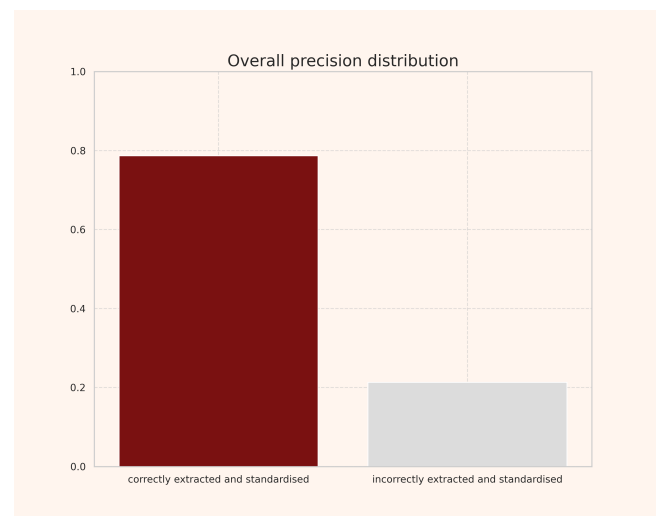


Figure 4: This figure presents the distribution of correctly and incorrectly extracted and standardized elements.

Figure 4 shows the distribution of the correctly extracted and standardized and incorrectly extracted and standardized elements. The graph shows how SkiLLMo achieves 79% overall precision in Skill Extraction and Standardization.

5.1 Discussion

In Figure 2 we can see that the retrieved skills are considered relevant by experts 91% of the times, indicating that the model is able to extract skills that are semantically related to the occupation.

In Figure 3 we can see that experts consider 80% of the extracted skills to be well mapping to ESCO skills. This indicates that 80% of the skills extracted by the LLM are semantically rich enough, leading to successful standardization. However, in some cases, a few skills may require further refinement in expression to ensure accurate standardization.

Finally, Figure 4 allows us to see the percentage of skills that are both extracted and standardised correctly. SkiLLMo totals an overall combined accuracy of 79%, which means that of the 91% of skills correctly extracted within the filtered OJAs, 79% of these are semantically rich enough to be correctly standardized on the ESCO taxonomy.

Table 2 shows an overview of the different precision performances in the skill extraction task. Despite no specific training for the skill extraction task, SkiLLMo achieves precision equal to that of the state of the art, thus testifying to the effectiveness of the tool. Despite the lack of specific model adaptation, the performance of the LLM in the skill extraction task has been notably positive, making it a promising approach for further exploration in future research.

Table 2: The table compares precision scores reported by various works on different benchmark datasets, highlighting the progress in skill extraction methods across diverse evaluation settings.

Work	Methodology	Precision
Barducci et al. [2]	BERT (NER)	62%
Gugnani et al. [13]	Exact Matching	80%
Javed et al. [17]	Exact Matching	90%
Bhola et al. [3]	BERT (XMLC)	90%
Jiechieu and Tsopze [18]	CNNs	91%
SkiLLMo	LLMs	91%

6 Conclusion and Future Developments

We propose SkiLLMo, a tool based on open source Transformer encoder and decoder models to extract skills from OJA and standardize these on the ESCO taxonomy. The use of an LLM, pre-trained to carry out different tasks, allows us to overcome the limit represented by the absence of a gold standard which does not allow training for the specific task. We enhance this process through a semantic search in a vector database representing the labels of the ESCO skills, in order to normalize the entities extracted by the LLM. We test this tool on 400 preprocessed and filtered OJAs, using a BERT model developed by us, obtaining an accuracy of 91% for the skill extraction step, an accuracy of 80% for the skill standardization step and a total accuracy combined of 79%.

The absence of a labeled Benchmark for the skill extraction task represents a limitation, as it does not allow an effective comparison between the different methodologies on the same dataset. Furthermore, our evaluation conducted by domain experts does not allow us to evaluate the completeness of finding the results with respect to the entire search space, not allowing us to quantify the recall of the system with respect to the skill extraction task. Moreover, given the size of the model used, a further step to investigate would be to optimise the computational resources required by trying to replicate the same results with smaller open source models. This would also allow a hypothetical real-time application of this tool. Differently, for the skill standardisation step, it is necessary to enrich the procedure in order to improve performance. One possibility consists in the semantic enrichment of the entity previously extracted by the LLM. This semantic and contextual enrichment with respect to the input OJA could allow the skill representation to approach correct representations of the corresponding ESCO skills, improving the performance of the standardization process.

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